Indices, Surds & Quadratics

Section 1: Laws of Indices & Surd Arithmetic

1 Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ completely.

a. $\frac{2x^2}{3y}$

b. $\frac{4x^2}{9y}$

c. $\frac{2x^3}{3y}$

d. $\frac{8x^2}{27y}$

2 Simplify $\sqrt{48} - \sqrt{12} + \sqrt{75}$ into the form $k\sqrt{3}$.

a. $7\sqrt{3}$

b. $5\sqrt{3}$

c. $9\sqrt{3}$

d. $6\sqrt{3}$

3 Rationalize the denominator of $\frac{6}{3-\sqrt{3}}$ and write in simplest form.

a. $3+\sqrt{3}$

b. $2+\sqrt{3}$

c. $3-\sqrt{3}$

d. $6+2\sqrt{3}$

Section 2: Completing the Square & Turning Points

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4 Express $2x^2 - 12x + 11$ in the form $a(x - h)^2 + k$.

a. $2(x - 3)^2 - 7$

b. $2(x - 3)^2 + 11$

c. $2(x - 6)^2 - 25$

d. $2(x - 3)^2 - 11$

5 Find the minimum value of $f(x) = 3x^2 + 18x + 20$ and the value of x at which it occurs.

a. Minimum is -7 at $x = -3$

b. Minimum is 20 at $x = 0$

c. Minimum is -7 at $x = 3$

d. Minimum is 7 at $x = -3$

6 The curve $y = -(x + 4)^2 + 15$ has a stationary point. State its nature and coordinates.

a. Maximum at $(-4, 15)$

b. Minimum at $(-4, 15)$

c. Maximum at $(4, 15)$

d. Minimum at $(4, -15)$

Section 3: The Discriminant & Nature of Roots

7 Find the set of values of k for which the equation $x^2 + kx + (k + 3) = 0$ has two distinct real roots.

a. $k < -2$ or $k > 6$

b. $-2 < k < 6$

c. $k \leq -2$ or $k \geq 6$

d. $k > 6$

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8 The line $y = 2x + c$ is tangent to the curve $y = x^2 - 4x + 7$. Find the exact value of c.

a. -2

b. 2

c. 7

d. -6

9 Determine the condition on p such that the quadratic $px^2 + 4x + p = 0$ has no real roots, where $p > 0$.

a. $p > 2$

b. $0 < p < 2$

c. $p > 4$

d. $p < -2$

Section 4: Disguised Quadratics & Exam-Style Modeling

10 Solve the disguised quadratic equation $x^4 - 5x^2 - 36 = 0$ for all real solutions.

a. $x = \pm 3$

b. $x = \pm 2$

c. $x = \pm 3, \pm 2i$

d. $x = 9$

11 Solve the equation $2^{2x} - 10(2^x) + 16 = 0$.

a. $x = 1$ or $x = 3$

b. $x = 2$ or $x = 8$

c. $x = 0$ or $x = 4$

d. $x = -1$ or $x = 3$

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12 A rectangular enclosure is formed using 80 meters of fencing against a straight stone wall. What is the maximum possible area of the enclosure in square meters?

a. $800\ m^2$

b. $400\ m^2$

c. $1600\ m^2$

d. $600\ m^2$

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